

Course Code: CS1108		
Name of the Programme: BCA (Hons.)		
Name of the Course: Linux and Shell Programming		
Semester: I		
Course Credit	No. of Hours Per Week (L+T+ P)	Total No. of Teaching Hours
4	3:0:2	75
Course Objectives:		
a) Provide a comprehensive understanding of the Linux operating system, its architecture, and core functionalities.		
b) Develop proficiency in using Linux command-line utilities, text-processing tools, and system operations.		
c) Enable students to write, debug, and execute shell scripts for task automation.		
d) Equip students with practical skills in system administration and routine Linux-based automation tasks.		

Course Outline (Syllabus of the Course)

Module - 1 : Introduction to Linux and the Command Line	No. of Hours
	9 Hrs.
History and Philosophy of Linux, Introduction to Open Source Software and Linux Distributions (e.g., Ubuntu, Fedora, CentOS), Linux System Architecture: Kernel, Shell, File System, Basics of Command Line Interface (CLI), Installing Linux (Ubuntu Server) on VirtualBox , Basic Navigation in Linux Terminal: cd, ls, pwd, mkdir, rm, cp, mv, Introduction to Text Editors: vi/vim	
Module - 2 : Core Linux Commands	No. of Hours
	9 Hrs.
Understanding the PATH Variable, Essential Commands: man, echo, printf, passwd, uname, who, date, File Handling Commands: cat, more, wc, lp, od, tar, gzip, File and Directory Management: pwd, cd, mkdir, rmdir, ls, cp, mv, rm, Security with File Permissions, Process and Disk Management: ps, w, ulimit, du, df, mount, umount, Networking Commands: arp, ftp, telnet, rlogin, Searching and Utilities: find, unlink, unmask	
Module - 3 : Text Processing and Advanced Utilities	No. of Hours

	9 Hrs.
Text Processing Tools: head, tail, sort, nl, uniq, grep (grep, egrep, fgrep), cut, paste, join, tee, File Comparison: comm, cmp, diff, Data Manipulation: tr, awk, sed, grep and sed: Usage, Scripts, Operations, Addresses, and Applications, Backup Utilities: cpio, tar, gzip	
Module - 4 : Shell Programming, Basics of Shell Scripting	No. of Hours
	9 Hrs.
Writing Shell Scripts, Using read and Command Line Arguments, Variables, Logical, and Conditional Operators (if, case), Computations with expr and String Handling, Loops: while, for, Advanced Features: set, shift, trap, and Debugging Shell Scripts, Input Validation and Data Entry Scripts, Introduction to Arrays and Functions in Shell	
Module - 5 : Advanced awk Scripting, Overview of awk	No. of Hours
	9 Hrs.
Structure, Execution, and Comments, Operators, Fields, and Built-in Variables, Output Formatting: print, printf, Escape Sequences, OFS Variable, User-Defined Variables and Parameter Passing, Control Structures: if-else, while, for Loops, Arrays: Declaration, Subscripts, Deleting Elements, User-Defined Functions and the getline Function, Regular Expressions and Advanced Use Cases	

Course Outcomes: After completing the course, the students will be able to: -
a) Understand the fundamental concepts and architecture of the Linux operating system.
b) Apply Linux commands to manage files, processes, and system operations effectively.
c) Develop and debug shell scripts to automate tasks within a Linux environment.
d) Analyze and process data using Linux text-processing tools such as grep , sed , and awk to solve real-world problems.

Textbooks

1	Nemeth, E., Snyder, G., Hein, T. R., Whaley, B., & Mackin, D. (2017). <i>UNIX and Linux system administration handbook</i> (5th ed.). Addison-Wesley Professional. ISBN: 978-0134277554
2	Das, S. (2012). <i>Unix concepts and applications</i> (4th ed.). McGraw Hill. ISBN: 978-0-07-063546-3

Reference Books

1	Afzal, I. (2019). <i>A complete course on Linux bash shell scripting with real life examples</i> . Packt Publishing. ISBN: 978-1838984083
2	Rosen, K., Host, D., & Klee, R. (2007). <i>The complete reference UNIX</i> (2nd ed.). Tata McGraw-Hill. ISBN: 0-07-065836-6
3	Blum, R., & Bresnahan, C. (2015). <i>Linux command line and shell scripting bible</i> (3rd ed.). Wiley. ISBN: 978-1118983843

Reference Links

1	https://onlinecourses.swayam2.ac.in/aic20_sp05/preview
2	https://www.coursera.org/learn/hands-on-introduction-to-linux-commands-and-shell-scripting

Laboratory Component

30 Hrs.

1	Practice basic Linux navigation commands: cd, ls, pwd, mkdir, rm, cp, and mv.
2	Edit and save files using text editors like vi and vim.
3	Explore and understand the PATH variable and execute essential commands like man, echo, passwd, and uname.
4	Perform file handling tasks using commands like cat, more, wc, lp, od, and tar.
5	Compress and extract files using gzip, tar, and cpio.

6	Manage files and directories using commands such as pwd, cd, mkdir, rmdir, ls, cp, mv, and rm.
7	Monitor system processes and disk usage with ps, w, df, du, mount, and umount.
8	Use networking commands like arp, ftp, telnet, and rlogin to explore basic networking tasks.
9	Search for files and directories using commands like find, grep, and unmask.
10	Process text files using tools like head, tail, sort, nl, uniq, and cut.
11	Compare files using commands like comm, cmp, and diff, and manipulate data with tr, paste, and join.
12	Create and debug shell scripts that include variables, loops, and conditional statements (if, case, while, for).
13	Implement advanced shell scripting features like set, shift, trap, and arrays.
14	Write awk scripts to process and format data, exploring control structures (if-else, while, for) and arrays.
15	Use regular expressions with awk and sed to perform pattern matching and text processing.
16	Develop backup scripts using tar, cpio, and gzip to automate data archiving and recovery.